SOURCE: MEPC.376(80).pdf
intensity of marine fuels (LCA Guidelines),
1 ADOPTS the Guidelines on life cycle GHG intensity of marine fuels (LCA Guidelines),
as set out in the annex to the present resolution;
2 AGREES that any regulatory application and implications of the LCA Guidelines
should be determined by the Committee in the process of developing regulatory provisions,
3 REQUESTS Member Governments to bring the annexed Guidelines to the attention
of shipowners, ship operators, shipbuilders, ship designers, energy companies, fuel producers,
bunkering companies, engine manufacturers and any other interested parties;
4 AGREES to keep these Guidelines under review in light of experience gained with
their implementation.
RESOLUTION MEPC.376(80) (adopted on 7 July 2023)
GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS (LCA GUIDELINES) MEPC 80/17/Add.1
Annex 14, page 2

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GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS

(LCA Guidelines)

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PART I: GENERAL

1 INTRODUCTION

1.1 These guidelines provide guidance on life cycle GHG intensity assessment for all fuels

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PART I: GENERAL

1 INTRODUCTION

1.1 These guidelines provide guidance on life cycle GHG intensity assessment for all fuels and other energy carriers (e.g. electricity) used on board a ship. These guidelines aim at covering the whole fuel life cycle (with specific boundaries), from feedstock extraction/cultivation/ recovery, feedstock conversion to a fuel product, transportation as well as distribution/bunkering, and fuel utiliza tion on board a ship. These guidelines also specify sustainability themes/aspects for marine fuels and define a Fuel Lifecycle Label (FLL), which carries information about fuel type, feedstock (feedstock type and feedstock nature/carbon source), conversion/production process (process type and energy used in the process), GHG emission factors, information on fuel blends and sustainability themes/aspects.
These guidelines specify the elements of FLL subject to verification/certification and include a general procedure on how the certification scheme/standards could be identified.

2 SCOPE

2.1 The scope of these guidelines is to address well-to-tank (WtT), tank-to wake (TtW),

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general procedure on how the certification scheme/standards could be identified.

2 SCOPE

2.1 The scope of these guidelines is to address well-to-tank (WtT), tank-to wake (TtW), and well-to-wake (WtW) greenhouse gases (GHG) intensity and sustainability themes/aspects related to marine fuels/energy carriers (e.g. electricity for shore power) used for ship propulsion and power generation onboard. The relevant GHGs included are carbon dioxide (CO 2), methane (CH4) and nitrous oxide (N2O). These guidelines are not intended to provide guidance for a complete IMO GHG inventory for international shipping. Emissions from cargo (e.g. volatile organic compounds (VOC)), or use of refrigerants are not included; other short -lived climate forcers and precursors such as non -methane volatile organic compounds (NMVOC), sulphur oxides (SO x), carbon monoxide (CO), particulate matter (PM) and Black Carbon are not part of the scope of these LCA guidelines.

2.2 The system boundaries of the WtW GHG emission factors calculation, in the context of these guidelines span the life cycle of fuels from their sourcing to production, conversion,

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not part of the scope of these LCA guidelines.

2.2 The system boundaries of the WtW GHG emission factors calculation, in the context of these guidelines span the life cycle of fuels from their sourcing to production, conversion, transport, distribution, and eventually their use on board ships based on an attributional approach.1 The possibility to expand the system boundaries for specific pathways in which the

feedstock is displaced from present use(s) will be assessed on a case-by-case basis.2 As such, emissions associated with the following life cycle stages of the fuel life cycle chain will be accounted for:

.1 feedstock extraction/cultivation/acquisition/recovery;
.2 feedstock (early) processing/ transformation at source;
.3 feedstock transport to conversion site;
.4 feedstock conversion to product fuel;
.5 product fuel transport/storage/delivery/retail storage/bunkering; and
.6 fuel utilization on board a ship.

2.3 Consistently with the attributional approach and using best available scientific evidence, the WtT emissions calculations (i.e. emissions related to the fuel sourcing,

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.6 fuel utilization on board a ship.

2.3 Consistently with the attributional approach and using best available scientific evidence, the WtT emissions calculations (i.e. emissions related to the fuel sourcing,
production, conversion, transport and delivery) are assessed regardless of the final use of